FLIP- FLOPS

There are many applications in which digital outputs are required to be generated in accordance with the sequence in which input signals are received. This requirement cannot be satisfied using combinational logic system. These applications require outputs to be generated that are not only dependent on the present input conditions but they also depend upon the past history of these inputs. The past history is provided by the feedback from the output back to the input.

The information stored in the memory elements at any given time defines the present state of the sequential circuit. The present state and external inputs determine the outputs and next state of sequential circuit. Thus we can specify the sequential circuits by the time sequence of external inputs, internal states and outputs.

sequential circuits are slower than combinational circuits and comparatively harder to design. Sequential circuits are classified depending on the timing of their signals : Synchronous sequential circuits and Asynchronous sequential circuits.

Latches Vs Flip Flops:

Latches and Flip flop are basic building blocks of the sequential circuits. Latches are controlled by enable signal and they are level triggered. Simple latch form the basis of flip flop. Each flip flop can store one bit of data. Flip flops are pulse or clock edge triggered .

Level Triggering

Positive level triggered

When a flip flop is required to respond at its HIGH state, a HIGH level triggering method is used. It is mainly identified from the straight lead from the clock input. Take a look at the symbolic representation shown below.

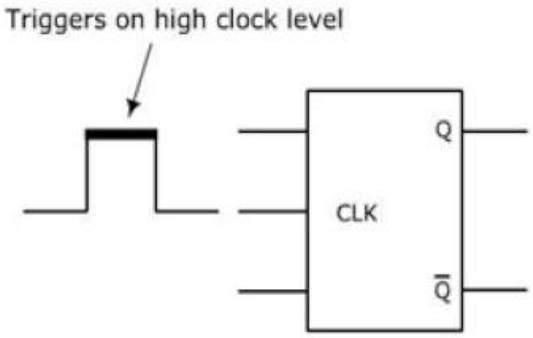


Figure 4.6 High Level Triggering

Negative level triggered :

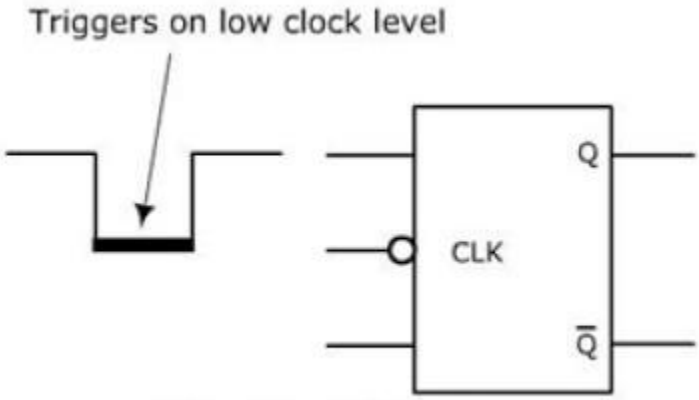
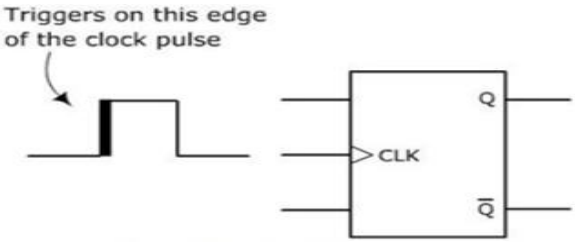


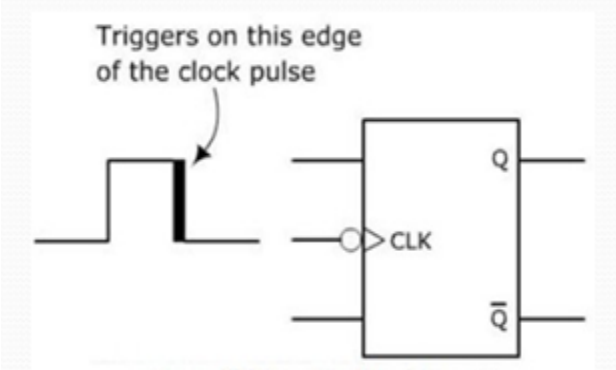
Figure 4.7 Low Level Triggering

Edge Triggering:

Positive Edge Triggering:



Negative Edge triggering:



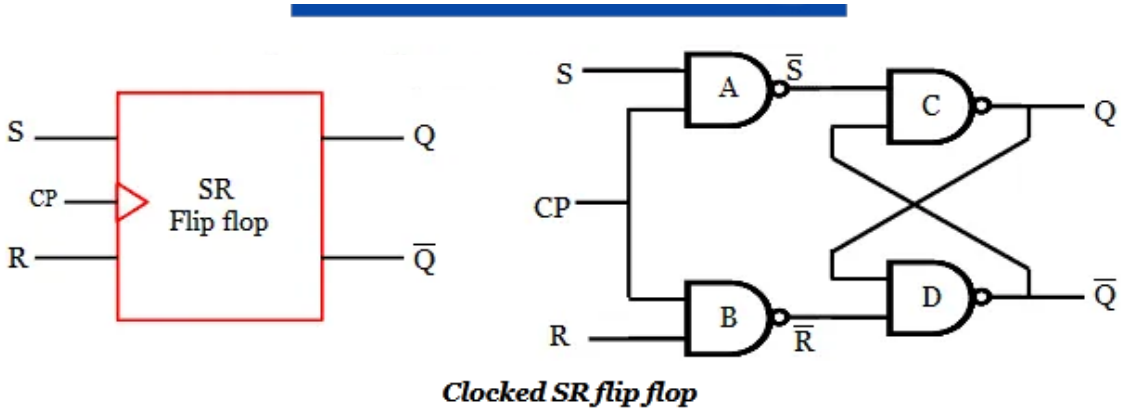
Types of Flip Flops

1. S-R flip flop
2. D flip flop
3. J-K flip flop
4. T flip flop

Clocked S-R flip flop:

The Clocked SR flip flop consists of 4 NAND gates, two inputs(S and R) and two outputs(Q and Q'). The clock pulse is given at the inputs of gate A and B.

If the clock pulse input is replaced by an enable input, then it is said to be SR latch. Let us assume that this flip flop works under positive edge triggering. The following figure shows the block diagram and the logic circuit of a clocked SR flip flop.



Clocked SR flip flop

Clk	S	R	Qn+1
1	0	0	Qn (No change)
1	0	1	0 (RESET)
1	1	0	1 (SET)
1	1	1	Inderminate

Working :

1)No Change state[S = 0, R = 0] :

When the clock pulse is applied, the output of NAND gates A and B will be S' = 1, R' = 1.

For this case, if Q = 0, Q' = 1, then both the inputs for NAND gate C are 1 and the output thus produced by gate C is Q_{+1} =0. The present state output is Q = 0 and the next state output is Q' = 0. Thus the state has no change. For the same SR inputs, if Q = 1, Q' = 0, the inputs for NAND

gate C will be 0 and 1. For these inputs, the output produced by the NAND gate is $Q_{+1} = 1$, hence there is no change in the state.

RESET state[S = 0, R = 1]:

Upon the application of the clock pulse, the output of NAND gate A and B are S' = 1, R' = 0.

Let the present state output be Q = 0 or Q = 1. For any of these inputs at the NAND gate D, the next state output produced is $Q'_{+1} = 1$. Now, the tw0 inputs for NAND gate C are S' = 1, Q' = 1, which produces an output at next state as $Q_{+1} = 0$.

For this case, whether the present state is either 0 or 1, it will produce an output 0, which will RESET the flip flop.

SET state[S = 1, R = 0]

When the clock pulse is applied, the output from the NAND gate A and B are S' = 0, R' = 1.

For this condition, irrespective of the present state input Q', the next state output produced by the NAND gate C is $Q_{+1} = 1$. Thus the two inputs of NAND gate D are R' = 1 and Q = 1, which produces an output $Q'_{+1} = 0$. So, in this case, whether the present state output is either 0 or 1, the next state output is logic 1, which will SET the flip flop.

Indeterminate or Invalid state[S = 1, R = 1]

For the inputs S = 1 and R = 1, the NAND gates A and B produces the output S' = 0, R' = 0.

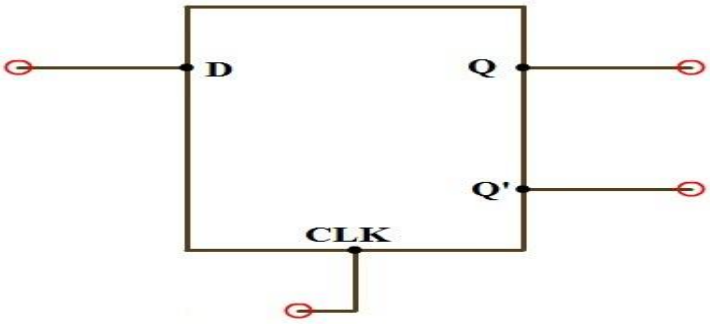
Now, if Q = 0 and Q' = 1, the inputs for NAND gate C will be S' = 0 and Q' = 1. The output produced from NAND gate C is $Q_{+1} = 1$. Similarly, the two inputs for NAND gate D will be R' = 0 and Q = 0. The output produced from the NAND gate D is $Q'_{+1} = 1$.

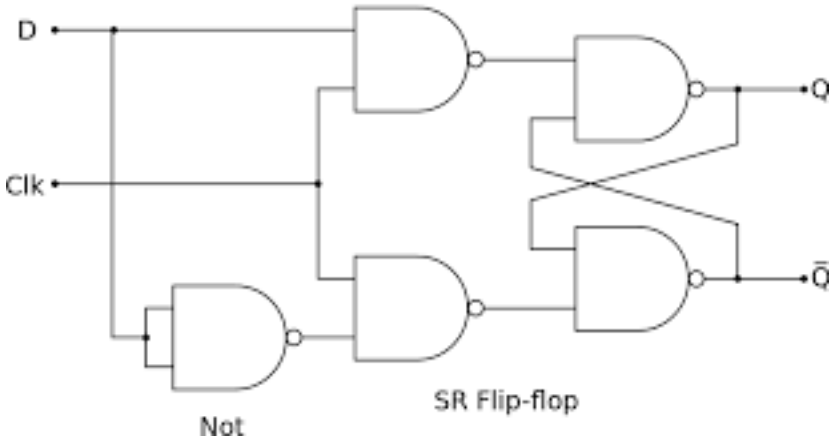
If Q = 1 and Q' = 0, the output produced from the NAND gate C is Q_{+1} = 1 for the inputs S' = 0 and Q' = 0. For the same value of Q and Q', output produced from NAND gate D is Q'_{+1} = 1, where the inputs are R' = 0 and Q = 1.

For this case, it is observed that the next state output Q_{+1} = 1 and Q'_{+1} = 1. This is an impossible output because Q and Q' should be complement with each other. So it is an indeterminate or invalid state.

Clocked D flip flop:

In D flip flop basic S-R flip flop is used with complemented inputs. Logic symbol, circuit diagram and truth table for positive edge triggered D flip flop is shown.



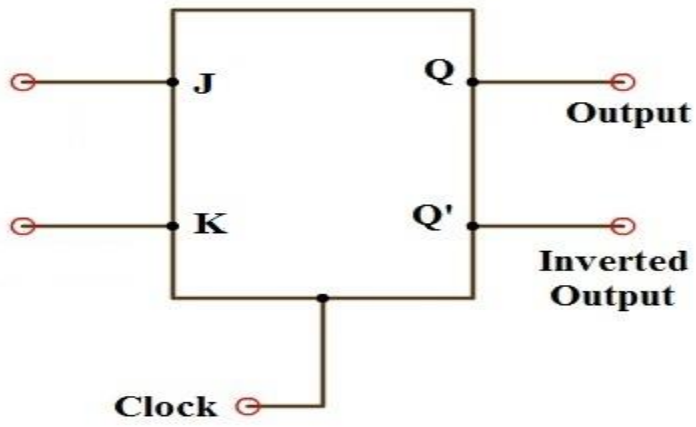


CP	D	Q_{+1}
1	0	0
1	1	1
0	x	Q_n

As shown in the above fig. circuit output responds to the D input only at the positive edges of the clock pulse. At any other instants of time the D flip flop will not respond to the changes in input. Looking at the truth table, we can realise that Qn+1 function follows D input at the positive going edge of clock pulses. Hence characteristics equation for D flip flop is Qn+1 =D. however output is delayed by one clock cycle. Hence D flip flop is called as delay flip flop.

Clocked J-K flip flop:

The uncertainity in the state of an S-R flip flop when S=R=1 can be eliminated by converting it into J-K flip flop. The data inputs are J and k which are ANDed with Q' and Q repectively, to obtain S R inputs as shown in fig. Thus S= J.Q' and R=K.Q



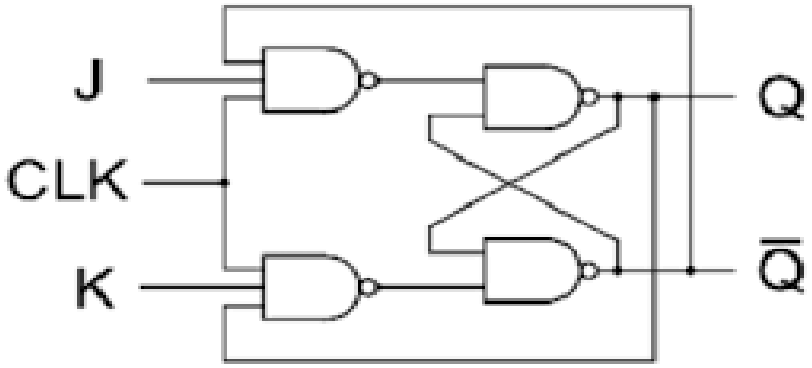


Figure 4: JK Flip Flop

CP	J	K	Q_{+1}	State
1	0	0	Q_n	NO CHANGE
1	0	1	0	RESET
1	1	0	1	SET
1	1	1	$\overline{Q}_n$	TOGGLES

Working:

case 1: J=K=0

when J=K=0,S=R=0 and according to truth table of S-R flip flop there is no change in the output.

Case 2: J=1 ,K=0

Consider Q=0,Q'=1: when J=1 ,K=0 and Q=0, then S=1 and R=0, according to truth table of S-R flip flop it it set condition and output Q=1

Consider Q=1,Q'=0: when J=1 ,K=0 and Q=1, then S=0 and R=0, according to truth table of S-R flip flop there is no change in the output and therefore Q=1 and Q'=0.

Case 3: J=0,K=1

Consider Q=0,Q'=1: when J=0 ,K=1 and Q=0, then S=0 and R=0, according to truth table of S-R flip flop there is no change in the output and Q=0, Q'=1.

Consider Q=1,Q'=0: when J=0 ,K=1 and Q=1, then S=0 and R=1, according to truth table of S-R flip flop it is reset state and output Q=0.

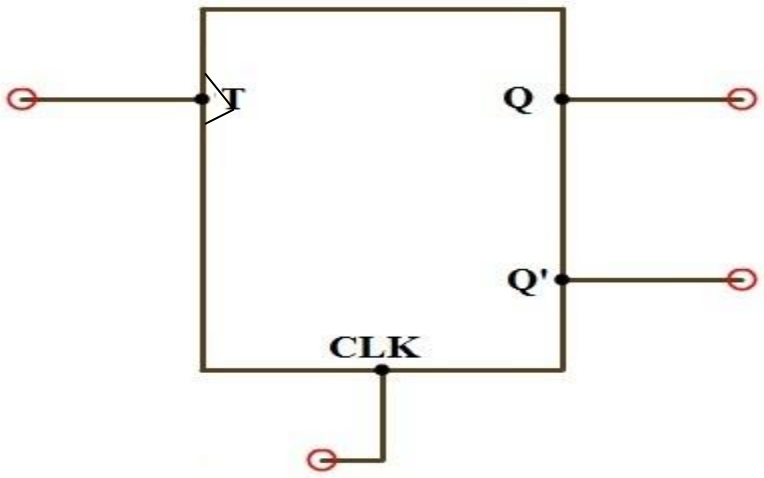
Case 4: J=1,K=1

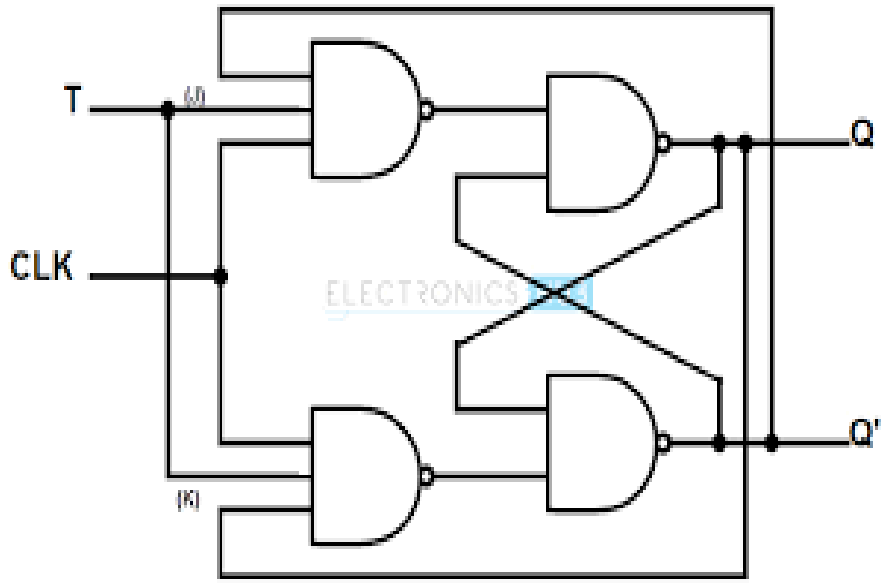
Consider Q=0,Q'=1: when J=1 ,K=1 and Q=0, then S=1 and R=0, according to truth table of S-R flip flop it is set state and Q=1

Consider Q=1,Q'=0: when J=1 ,K=1 and Q=1, then S=0 and R=1, according to truth table of S-R flip flop it is reset state and output Q=0.Thus input J=K=1 **toggles** the flip flop output.

Clocked T flip flop

T flip flop is also known as toggle flip flop. It is modification of J K flip flop. As shown in the figure T flip flop is obtained from JK flip flop by connecting both inputs J AND K together.





T	Present state Q_n	Next state Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Working:

When T=0, J=K=0 and hence there is no change in the output.

When T=1, J=K=1, and hence output toggles.

Applications of Flip flops

1. Counters

2.Registers

3. As memory Element

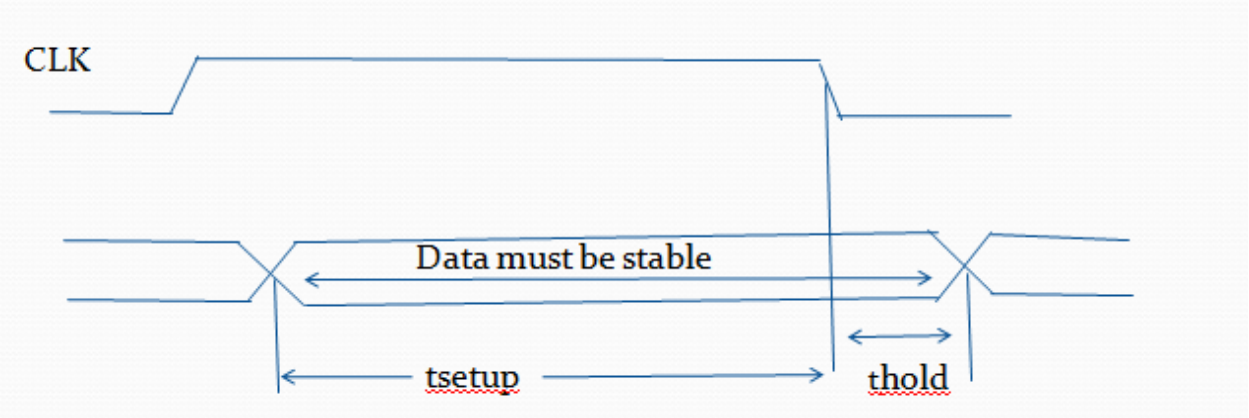
4. Data Transfer

Flip Flop Characteristics

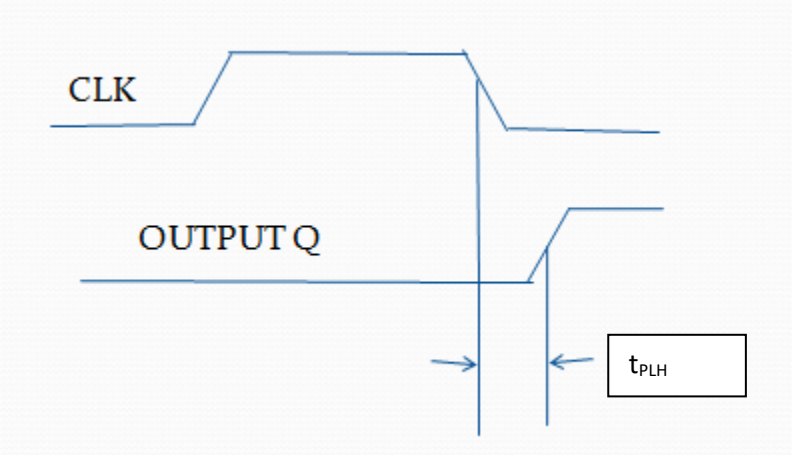
- Set up time
- Hold time
- Propagation delay(t_{PLH})
- Propagation delay(t_{PHL})
- Maximum frequency

Setup time : Minimum time required to maintain constant voltage levels at excitation inputs of the flip flop prior to the triggering edge of clock pulse in order for the levels to be reliably clocked into the flip flop.

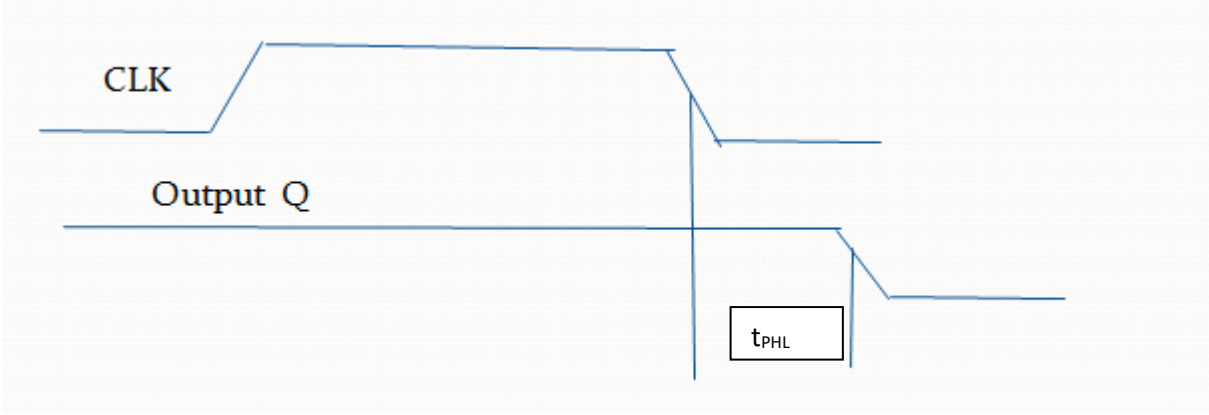
Hold time: Minimum time for which voltage levels at excitation inputs must remain constant after triggering edge of clock pulse in order for the levels to be reliably clocked into the flip flop .



- **Propagation Delay** : Time required to change output after application of input.
- **Propagation Delay** (t_{PLH}): It is measured from the triggering edge of the clock pulse to the low to high transition of the output.



Propagation delay (t_{PHL}): It is measured from the triggering edge of the clock pulse to the high to low transition of the output.



- Maximum frequency : highest frequency of clock pulse for proper and stable operation